Projected Toroidal Geometry and Testable Angular Statistics for Transient Stellar Visibility

A Conditional Search Framework for the Vera C. Rubin Observatory

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SANDBOXED: Geometry and statistical forecasts only; no detection or derived cosmological metric

Abstract

We formulate a falsifiable observational search for spatial and temporal organization among sources classified as transiently vanished or reappeared in repeated astronomical imaging. The motivating H(s)H "Cruller" conjecture places a cosmological origin locus on an inner toroidal longitude and proposes a gravitationally induced deformation of observable null-path structure. The dynamical metric and gravitational-polarization constitutive law remain unknown. We therefore separate (i) exact Euclidean projection of a model inner ring, (ii) a conventional tidal-shear lens that illustrates changing image multiplicity, and (iii) survey-mask-conditioned angular statistics. We derive the off-axis projected sky curve and give a closed-form caustic threshold for a point lens plus quadrupolar shear. A numerical sensitivity example illustrates how a pre-registered angular template could be evaluated; it does not forecast actual event counts. Notably, an image pair's disappearance does not imply loss of all images of a stellar source. We specify the missing requirements for a physical prediction and the null tests needed to distinguish geometric signals from survey artifacts.

Keywords: gravitational lensing; caustics; celestial sphere; toroidal geometry; time-domain astronomy; survey selection function; LSST.

1 Scope and epistemic status

Our question is observational: if an evolving propagation geometry modulates stellar visibility, do event positions and epochs exhibit large-scale structure? A physically predicted pattern must be computed *before* examining its candidate catalog. This note supplies an explicitly parameterized *template family*, not a demonstrated gravitational effect or a prediction of sky coordinates. The physical hypothesis involves an inward-directed, left-biased worldtube geometry with finite-core resistance and a possible gravitational polarization response. Neither a Lorentzian metric nor its stress-energy source has yet been derived. We do not equate geometric visibility modulation with photon-to-neutrino conversion, cosmic birefringence, or known transient-star populations.

2 Geometry: toroidal inner circumference

Consider only an illustrative three-dimensional torus section

$$X(u,v) = ((R + r\cos v)\cos u, (R + r\cos v)\sin u, r\sin v), \quad R > r > 0. \tag{1}$$

Its inner longitude is at $v = \pi$, with radius $a = R - r$ and length $L_{\text{in}} = 2\pi a$; the outer longitude has length $L_{\text{out}} = 2\pi(R + r)$. The section's induced metric and intrinsic Gaussian curvature are

$$ds^2 = (R + r\cos v)^2 du^2 + r^2 dv^2, \qquad K(v) = \frac{\cos v}{r(R + r\cos v)}. \tag{2}$$

In particular $K(\pi) = -1/[r(R-r)]$. This is *surface* curvature, not automatically the spacetime Ricci tensor or a gravitational force. A conditional circumferential stretch restriction $\Lambda_{\text{geom}} = (R+r)/(R-r) \le \Lambda_{\max}$ would imply

$$R - r \ge \frac{2r}{\Lambda_{\max} - 1}. \tag{3}$$

A separately justified finite-core *longitudinal closure length* L_{core} would require $R - r \ge L_{\text{core}}/(2\pi)$. These are conditional geometric bounds; a transverse core thickness is not automatically a minimum longitudinal length.

3 Sky projection from an off-axis observer

Let $S(\phi) = (a\cos\phi, a\sin\phi, 0)$ parametrize the inner ring and place an observer at $O = (d, 0, z)$, with $z > 0$. Its Euclidean sightline direction is

$$\hat{n}(\phi) = \frac{S(\phi) - O}{\sqrt{a^2 + d^2 + z^2 - 2ad\cos\phi}}. \tag{4}$$

Relative to the torus symmetry axis, the apparent latitude is

$$b(\phi) = \arcsin\left[\frac{-z}{\sqrt{a^2 + d^2 + z^2 - 2ad\cos\phi}}\right]. \tag{5}$$

For $d = 0$, $b = -\arctan(z/a)$ is constant. For $0 < d < a$, its extrema are $-\arctan[z/(a-d)]$ and $-\arctan[z/(a+d)]$. Thus an inner ring yields a small circle only for an on-axis observer and generally a warped celestial curve for an off-axis observer. These are straight-line projections of a *section*, not GR null geodesics or direct caustic predictions.

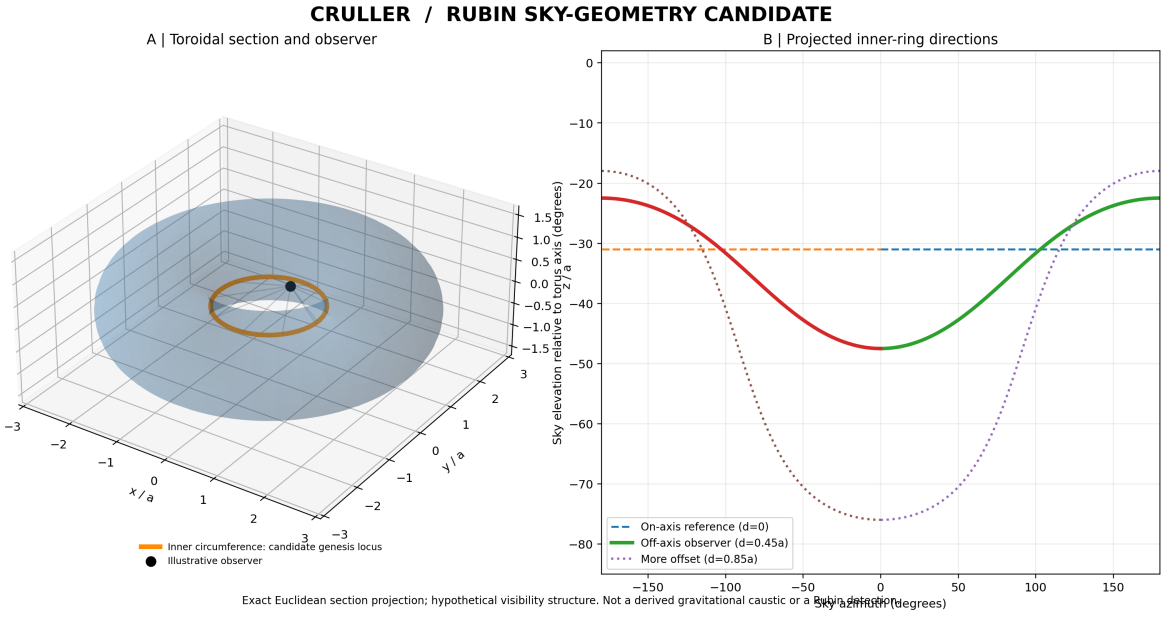


Figure 1: Illustrative toroidal-section and observer-dependent sky projection. Ring geometry and observer locations are chosen for visualization; the colored curves are not predicted disappearance-event contours.

4 Lensing control: an image fold is not a total blackout

A conventional point-mass lens of normalized strength $m > 0$ and constant external quadrupolar shear $0 < \gamma < 1$ obeys

$$\beta_x = (1-\gamma)x - \frac{mx}{x^2 + y^2}, \qquad \beta_y = (1+\gamma)y - \frac{my}{x^2 + y^2}. \tag{6}$$

For a source at $(b, 0)$, two axial image roots always exist:

$$x_\pm = \frac{b \pm \sqrt{b^2 + 4(1-\gamma)m}}{2(1-\gamma)}, \qquad y = 0. \tag{7}$$

A further pair, with $x = -b/(2\gamma)$ and

$$y_\pm = \pm\sqrt{\frac{m}{1+\gamma} - \frac{b^2}{4\gamma^2}}, \tag{8}$$

exists only if $|b| < b_c = 2\gamma\sqrt{m/(1+\gamma)}$. For $m = 1$ and $\gamma = 0.2$, the exact transition is at $b_c = 0.36514837\ldots$; the image count switches between four and two. The two permanent images constitute a direct counterexample to interpreting loss of a local fold pair as complete stellar disappearance. Near a caustic, point-source image magnification can instead become large; finite source size regularizes this behavior.

5 A survey-level observable and conditional sensitivity

Let $\hat{n}$ be the sightline to a candidate source. A generic narrow-band angular template can be written

$$|\arcsin(\hat{n} \cdot \hat{a}) - \delta_0| < w, \tag{9}$$

with unknown axis $\hat{a}$, offset δ_0, and half-width w. For full, uniformly sampled sky and $|\delta_0| + w < \pi/2$, its area fraction is $p = \cos\delta_0 \sin w$. For $\delta_0 = 25^\circ$ and $w = 2^\circ$, $p = 0.0316297$ or about 1305 deg^2. These are *assumed* angles chosen to demonstrate scale, not values obtained from H(s)H. With a specific survey selection mask, expected event counts must instead be based on eligible source exposure and detection efficiency. A conditional Poisson-rate formulation is

$$\lambda_i = N_{*,i}\,\epsilon_i\,[r_{0,i} + A\,K_i(\hat{a}, \delta_0, w, t)], \tag{10}$$

for pixel/epoch index i, eligible monitored sources $N_{*,i}$, recovery efficiency ϵ_i, empirically constrained background rate $r_{0,i}$, and precomputed template K_i. Parameters used to construct a candidate hypothesis must be frozen before testing on withheld data. A fitted axis, width, and event rate incur look-elsewhere penalties. A prior toy simulation assumed 1000 independent disappearance candidates, a southern-sky illustrative mask, and twice the rate inside a pre-fixed band. Under those assumptions, the expected count in the band increased from 41.9 to 80.5, giving a nominal fixed-test contrast $Z \simeq 6.08$. This is solely a *sensitivity illustration*; it is neither a Rubin forecast nor evidence for the hypothesis. The mask used was not the real Rubin footprint.

6 From section geometry to an actual physical forecast

The claim that gravitational polarization causes light-cone wrinkles requires a physically typed response, such as an induced stress-energy perturbation $\delta T_{\mu\nu}[P]$ satisfying the Einstein constraints and covariant conservation. From a consistent metric $g_{\mu\nu}[P]$, observer/source worldlines, and admissible global identifications, one must integrate the full null-geodesic boundary-value problem and the optical Jacobi map. Conjugate points satisfy $\det\mathcal{D} = 0$, but are not automatically regions of darkness. A genuine flux-based criterion is $F_{\rm obs}(\hat{n}, t) < F_{\rm lim}(\hat{n}, t)$, with $F_{\rm obs}$ summed over *all* accessible images, their delayed source luminosities, lens magnifications, and propagation losses. The model must independently predict visibility recovery. Frame dragging, optical birefringence, and any photoneutrino mechanism should be introduced only through separately derived transport rules. A static projected ring cannot encode left-handedness: signed time evolution is required.

7 Tests and failure criteria

The primary analysis would compare disappearance and confirmed reappearance candidates against exposure- and cadence-matched simulated catalogs. Conditions for physical interest include repeatable angular clustering after correction for plate footprints, crowding, extinction, proper motion, blending, crossmatch failure, and instrumental artifacts; coherent position/time structure not reconstructed from the same data; and predicted astrometric,

magnification, and polarization signatures. If a proposed metric predicts a specific moving caustic, it must also calculate whether the *total* flux declines rather than brightens. Null outcomes should bound the amplitude and angular coherence of the proposed response, not automatically refute unrelated SAT/H(s)H components.

8 Conclusions

We obtain a closed-form projected-sky locus for an off-axis view of an inner toroidal ring and an exact image-multiplicity threshold in a standard quadrupolar lens. The first establishes that apparent rings need not remain narrow small circles for off-axis observers. The second establishes that ordinary fold disappearance/return is insufficient to establish disappearance of a star. These calculations support a preregistered, survey-aware search for large-scale spatial and temporal structure in candidate source-visibility events. They do not establish a unique location, width, handedness, transient rate, or physical mechanism. The outstanding task is deriving the Cruller finite-core gravitational response, native Lorentzian metric, global null propagation, and all-image observed flux.

Source and prior-work note

The motivating inner-ring, finite-core and gravitational-polarization hypotheses come from the continuing SAT/H(s)H research correspondence, with internal dated source records in `WORKSPACES/MERIDIAN/` of https://github.com/Satobloc/HsH. The publication workflow follows `WORKSPACES/COMMON/LIVE_NEW_PAPERS_STANDARD.md`. External control references: Rubin Observatory, https://rubinobservatory.org/for-scientists/rubin-101/the-legacy-survey-of-space-and-time-lsst; VASCO, Villarroel et al., "The Vanishing & Appearing Sources during a Century of Observations Project: I", https://arxiv.org/abs/1911.05068. No independent observational data were analyzed for this manuscript.